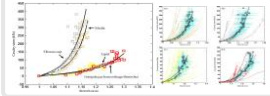


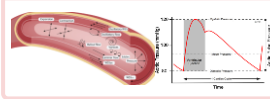
Step 1. Create data set

Input Parameters

Material properties



Hemodynamic

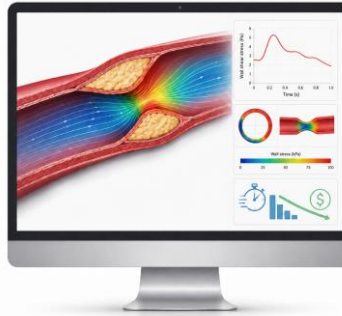


Morphology

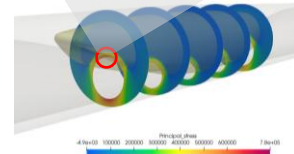
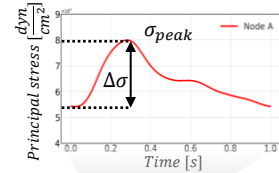


Computational Analysis

Computationally effective FSI simulation



Output Parameters



$$PSS \equiv \max_{i \in N} \sigma_{peak,i}, \Delta PSS \equiv \max_{i \in N} \Delta \sigma_i$$

Step 2. Rupture Vulnerability Index (VI) selection

$$VI \equiv \text{stress}/\text{strength} \\ \text{stress} \in \{PSS, \Delta PSS\}, \text{strength} \propto E_{fc}^\alpha, \alpha \in \{0.0, 0.5, 1.0\}$$

☐ $VI_{clinical}$: VI satisfies all 7 clinical high rupture risk plaque features

$$VI_1 = PSS/E_{fc}^{0.0}$$

$$VI_2 = PSS/E_{fc}^{0.5}$$

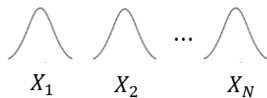
$$VI_3 = PSS/E_{fc}^{1.0}$$

$$VI_4 = \Delta PSS/E_{fc}^{0.0}$$

$$VI_5 = \Delta PSS/E_{fc}^{0.5}$$

$$VI_6 = \Delta PSS/E_{fc}^{1.0}$$

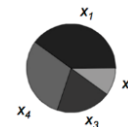
Step 3. Surrogate Model Training & Sensitivity Analysis



Input Parameters

$$VI_{clinical} = f(X_1, \dots, X_N)$$

Surrogate Model



Sensitivity Analysis